Day 5 : Network Testing Tools Theory

Task 1 : Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Command Used

ping www.youtube.com

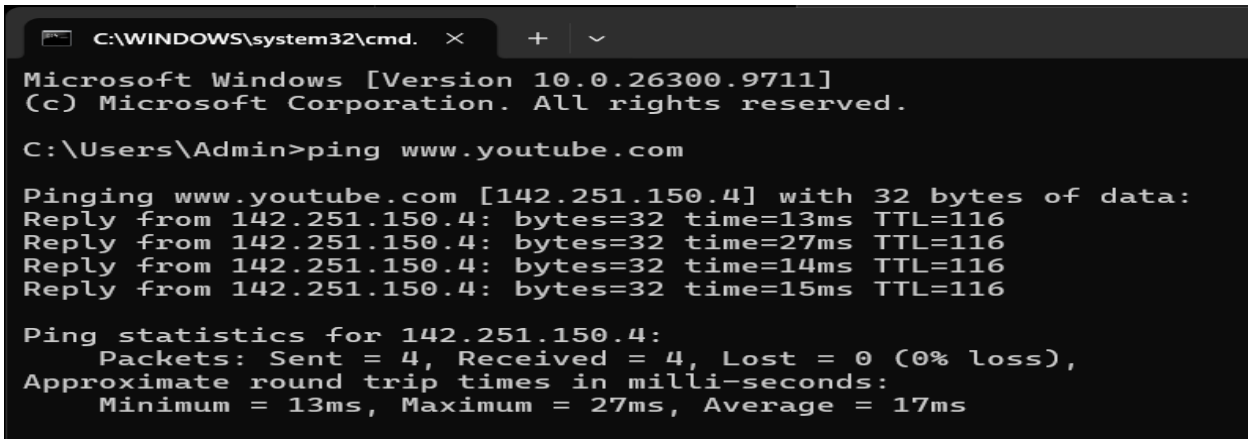
Test Results

Parameter	Actual Result
Destination	www.youtube.com
Resolved IP Address	142.251.150.4
Packets Sent	4
Packets Received	4
Packets Lost	0 (0% loss)
Minimum Response Time	13 ms
Maximum Response Time	27 ms
Average Response Time	17 ms
TTL	116

Explanation

I opened Windows Command Prompt and executed the command ping www.youtube.com. The hostname resolved to the IP address 142.251.150.4. I received replies to all four requests, and no packets were lost. The minimum response time was 13 ms, the maximum was 27 ms, and the average was 17 ms.

These results indicate that www.youtube.com was reachable from my computer during the test and that the connection had a low average response time.



Task 2 : Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Command Used

tracert www.flipkart.com

Results

The command resolved the destination to e348826.dsca.akamaiedge.net with IP address 23.55.244.187.

Hop	Hostname / IP Address	Response Times
1	192.168.0.1	4 ms, 2 ms, 6 ms
2	gpon.net (192.168.1.1)	2 ms, 1 ms, 1 ms
3	10.130.192.1	4 ms, 14 ms, 3 ms
4	10.111.13.5	34 ms, 116 ms, 72 ms
5	103.241.47.53	Timeout, Timeout, 67 ms
6	103.241.47.110	19 ms, 13 ms, 17 ms
7	ae3.gpx-bom2.netarch.akamai.com (104.70.119.207)	15 ms, 16 ms, 14 ms
8	a23-55-244-187.deploy.static.akamaitechnologies.com (23.55.244.187)	20 ms, 20 ms, 26 ms

Explanation

1. I opened Command Prompt on my Windows computer.
2. I executed tracert www.flipkart.com and waited for the trace to finish.
3. The command displayed eight hops before reaching the destination IP address, 23.55.244.187.
4. The first hop, 192.168.0.1, represents my local network gateway.
5. Some intermediate hops use private IP addresses, such as 10.130.192.1 and 10.111.13.5. These may belong to internal network infrastructure.

6. At hop 5, two probes timed out, but the third received a response. This indicates that some traceroute probes did not receive replies; it does not necessarily mean the network connection failed.
7. The final hop responded successfully, and the output displayed **“Trace complete.”**

```
C:\Users\Admin>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.55.244.187]
over a maximum of 30 hops:

  1     4 ms     2 ms     6 ms  192.168.0.1 [192.168.0.1]
  2     2 ms     1 ms     1 ms  gpon.net [192.168.1.1]
  3     4 ms    14 ms     3 ms  10.130.192.1 [10.130.192.1]
  4    34 ms   116 ms    72 ms  10.111.13.5 [10.111.13.5]
  5     *        *       67 ms  103.241.47.53
  6    19 ms    13 ms    17 ms  103.241.47.110
  7    15 ms    16 ms    14 ms  ae3.gpx-bom2.netarch.akamai.com [104.70.119.207]
  8    20 ms    20 ms    26 ms  a23-55-244-187.deploy.static.akamaitechnologies.com [23.55.244.187]

Trace complete.
```

Task 3 : Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Command Used

nslookup www.zomato.com

Results Obtained

Parameter	Result
Domain Name	www.zomato.com
DNS Server	192.168.0.1
DNS Server Address	192.168.0.1
Canonical Name	e70929.dsca.akamaiedge.net
IPv4 Address 1	23.55.244.186
IPv4 Address 2	23.55.244.185
IPv6 Address 1	2600:140f:1e00::1737
IPv6 Address 2	2600:140f:1e00::1737
Alias	zomato.edgekey.net

Explanation

1. I opened Windows Command Prompt.
2. I entered the command nslookup www.zomato.com.
3. The command showed 192.168.0.1 as the DNS server used for the lookup.
4. The result displayed a non-authoritative answer, which means the response was provided by a DNS resolver rather than directly by the domain's authoritative DNS server.
5. The domain resolved through the hostname e70929.dsca.akamaiedge.net and returned two IPv4 addresses and two IPv6 addresses.
6. The output also showed zomato.edgekey.net as an alias for www.zomato.com.

```
C:\Users\Admin>nslookup www.zomato.com
Server:  192.168.0.1
Address:  192.168.0.1

Non-authoritative answer:
Name:    e70929.dsca.akamaiedge.net
Addresses:  2600:140f:1e00::1737:f4d0
          2600:140f:1e00::1737:f4c3
          23.55.244.186
          23.55.244.185
Aliases:  www.zomato.com
          zomato.edgekey.net
```

Task 4 : Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Command Used

ping 192.168.0.103

Results Obtained

Parameter	**Result**
Destination IP Address	192.168.0.103
Packets Sent	4
Packets Received	4
Packets Lost	0 (0%)
Minimum Response Time	63 ms
Maximum Response Time	246 ms
Average Response Time	143 ms
TTL	64

Explanation

1. I opened Windows Command Prompt on my computer.
2. I executed ping 192.168.0.103 to test connectivity with another device on the local network.
3. The destination replied to all four ICMP Echo Requests.
4. No packets were lost, which indicates that the destination was reachable during the test.
5. The average response time was 143 ms, with response times ranging from 63 ms to 246 ms.

What Happens If the Device Is Offline?

If the destination device is switched off, disconnected from the network, or configured to block ICMP requests, the ping command may display **"Request timed out."** If the device is unreachable, all four requests may fail, resulting in 100% packet loss. However, a timeout alone does not prove that the device is offline because a firewall may block ping requests.

- **Screenshot**

```
C:\Users\Admin>ping 192.168.0.103

Pinging 192.168.0.103 with 32 bytes of data:
Reply from 192.168.0.103: bytes=32 time=63ms TTL=64
Reply from 192.168.0.103: bytes=32 time=198ms TTL=64
Reply from 192.168.0.103: bytes=32 time=65ms TTL=64
Reply from 192.168.0.103: bytes=32 time=246ms TTL=64

Ping statistics for 192.168.0.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 63ms, Maximum = 246ms, Average = 143ms
```

If Device Is Offline:

```
C:\Users\Admin>ping 192.168.0.103

Pinging 192.168.0.103 with 32 bytes of data:
Reply from 192.168.0.106: Destination host unreachable.
Reply from 192.168.0.106: Destination host unreachable.
Reply from 192.168.0.106: Destination host unreachable.
Reply from 192.168.0.106: Destination host unreachable.

Ping statistics for 192.168.0.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Task 5 : Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

- The **ping** tool is useful for checking connectivity and response time. In my test, ping www.youtube.com received all four replies with 0% packet loss and an average response time of 17 ms. For a slow connection, ping can help identify high or fluctuating response times; my local device test had an average of 143 ms, with a maximum of 246 ms. The **tracert** tool is useful for investigating the route taken by network traffic.
- My tracert www.flipkart.com test reached the destination in eight hops, although some probes at hop 5 timed out.
- The **nslookup** tool is useful for diagnosing DNS issues.
- My nslookup www.zomato.com test returned IPv4 and IPv6 addresses, showing that the domain resolved successfully.
- If a website does not load, I would first use ping to check reachability, tracert to investigate the route, and nslookup to verify DNS resolution.
- These tools provide different information and should be used together when necessary.